Thapar Institute of Engineering and Technology, Patiala
Electronics and communication Engineering (ECED)
Information and Communication Theory (UEC-310) Jan-June2026
Assignment

Date of submission: April 25, 2026 **Max. Marks**. 10

Assignment Statement

Reliable Data Transmission in a Satellite Communication System: A satellite communication company is transmitting environmental data from remote sensors on Earth to a ground station. Due to long-distance transmission and atmospheric disturbances, the communication channel is noisy, leading to possible data corruption.

To ensure efficient and reliable transmission, the system uses:

- **Source coding** to compress data
- **Channel coding** to detect and correct errors

The system transmits binary data sequences generated by sensors where:

- Probability of bit **'1'** = 0.7
- Probability of bit **'0'** = 0.3

A **Binary Symmetric Channel (BSC)** is used with a bit error probability of $\boldsymbol{p = 0.1}$.

Part A: Analytical Tasks

Q1. Entropy and Source Coding

- Compute the entropy of the source.
- What does this entropy represent in the context of data compression?
- Suggest whether compression is possible and justify.

Q2. Coding Efficiency

A coding scheme produces an average code length of **1.2 bits/symbol**.

- Calculate coding efficiency.
- Comment on optimality.

Q3. Channel Capacity

For a Binary Symmetric Channel:

- Compute channel capacity.
- Interpret the result in terms of maximum reliable transmission rate.

Part B: Simulation-Based Tasks

Activity 1: BSC Channel Simulation

Objective: Simulate transmission over a noisy channel and compute Bit Error Rate (BER).
Tasks:

1. Generate 10,000 random bits with given probabilities
2. Pass through BSC
3. Compute BER

Activity 2: Entropy vs Probability Plot

Objective: Analyze how entropy varies with probability.
Tasks:

1. Vary p from 0 to 1
2. Compute entropy for each value
3. Plot $H(p)$ vs p

Activity 3: Channel Capacity vs Noise

Objective: Study effect of noise on channel capacity.
Tasks:

1. Vary p from 0 to 0.5
2. Compute capacity
3. Plot capacity vs p

Deliverables:

- Submit a Jupyter notebook / Python script or Matlab code with your simulation code and plots.
- Provide a short PDF report (2–3 pages) explaining your approach, assumptions, analysis, and conclusions.
- Ensure your code is well-commented and reproducible.

Tools Allowed:

Python (NumPy, SciPy, Matplotlib, Seaborn), or MATLAB.